

Non Linear Regression - MTH 686

PROJECT REPORT

Swarnim Verma

Roll: 231071

Department of Mathematics and Scientific Computing, IIT Kanpur

November 11, 2025

Abstract

This report analyzes the supplied dataset (t_i, y_i) under three candidate models: (1) exponential sum, (2) rational function, and (3) polynomial model. For each, we derive the least-squares estimators, discuss the numerical estimation methods, and compare the results based on goodness-of-fit and statistical diagnostics. Confidence intervals are obtained using the Fisher Information Matrix. All computations are performed in Python (codes included in Appendix).

Contents

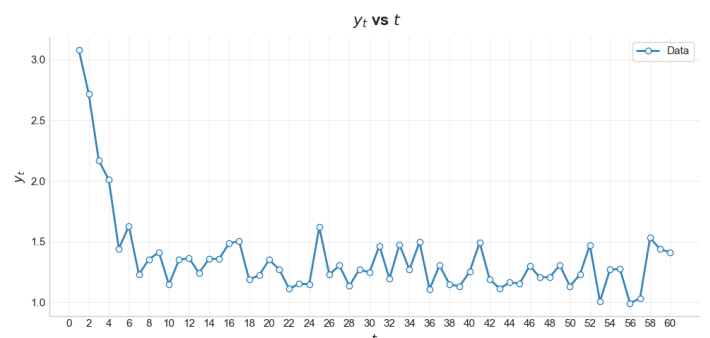
0 Dataset, Assumptions and Supplementary Mathematical Results	1
1 Model 1: Exponential Sum Model	2
1.1 Osborne's Method	2
1.1.1 Model parameter estimates	2
1.1.2 Estimate of σ^2 , Plotting of Residuals and Normality check	3
1.1.3 Fischer Information Matrix and 95 % Confidence Intervals	3
2 Model 2: Rational Model	4
2.1 Modified Newton Gauss Method (Levenberg–Marquardt Model)	4
2.1.1 Model Parameter estimates	4
2.1.2 Estimate of σ^2 , Plotting of Residuals and Normality check	4
2.1.3 Fisher Information Matrix and 95% Confidence Intervals	5
3 Model 3 : Polynomial Least Squares Fit (Degree 4)	5
3.1 Linear Regression	5
3.1.1 Model parameters estimates	5
3.1.2 Estimate of σ^2 , Plotting of Residuals and Normality check	5
3.1.3 Fischer Information Matrix and 95% confidence intervals	6
4 Conclusion	7
5 References	7

0 Dataset, Assumptions and Supplementary Mathematical Results

All the analysis that follows will be based on the **Dataset numbered 64**. We show the table below:

Table 1: Dataset: 64

t	y_t	t	y_t	t	y_t	t	y_t
1	3.0836	16	1.4858	31	1.4659	46	1.3009
2	2.7179	17	1.5034	32	1.1935	47	1.2051
3	2.1681	18	1.1869	33	1.4764	48	1.2064
4	2.0097	19	1.2220	34	1.2679	49	1.3062
5	1.4408	20	1.3511	35	1.4975	50	1.1323
6	1.6290	21	1.2701	36	1.1072	51	1.2274
7	1.2297	22	1.1104	37	1.3040	52	1.4714
8	1.3532	23	1.1524	38	1.1464	53	1.0066
9	1.4118	24	1.1479	39	1.1323	54	1.2683
10	1.1453	25	1.6204	40	1.2510	55	1.2746
11	1.3493	26	1.2273	41	1.4932	56	0.9919
12	1.3642	27	1.3037	42	1.1884	57	1.0317
13	1.2406	28	1.1352	43	1.1135	58	1.5305
14	1.3578	29	1.2692	44	1.1626	59	1.4370
15	1.3570	30	1.2468	45	1.1551	60	1.4112



Note: The assumption of normality is not essential for the numerical convergence of the Gauss–Newton or Osborne methods; however, it becomes crucial for statistical inference. Normality is required if one wishes to interpret the estimated parameters probabilistically—that is, to construct confidence intervals, perform hypothesis tests, or derive variance estimates. Therefore, in the subsequent analysis, we assume that the random errors are independently and normally distributed with mean zero and variance σ^2 .

Statistical Framework for Fischer Information Matrix and Confidence Interval Estimation

For a general nonlinear regression model $y_i = f(t_i, \beta) + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, the least-squares estimate $\hat{\beta}$ minimizes $S(\beta) = \sum_{i=1}^n [y_i - f(t_i, \beta)]^2$ and satisfies $\partial S(\beta)/\partial \beta = \mathbf{0}$. Since no closed-form solution exists, iterative algorithms such as Gauss–Newton, Levenberg–Marquardt, or Osborne’s method are used to approximate $\hat{\beta}$.

Let $\mathbf{J}$ denote the Jacobian matrix with elements $\mathbf{J}_{ij} = \frac{\partial f(t_i, \beta)}{\partial \beta_j} \big|_{\beta=\hat{\beta}}$. Using a first-order Taylor approximation $f(t_i, \beta) \approx f(t_i, \hat{\beta}) + \mathbf{J}_i(\beta - \hat{\beta})$, the covariance of $\hat{\beta}$ is estimated as $\text{Var}(\hat{\beta}) \approx \hat{\sigma}^2 (\mathbf{J}^\top \mathbf{J})^{-1}$, where $\hat{\sigma}^2 = \frac{\sum_{i=1}^n [y_i - f(t_i, \hat{\beta})]^2}{n-p}$. The Fisher Information Matrix is $\text{FIM} = \frac{1}{\sigma^2} \mathbf{J}^\top \mathbf{J}$, and the standard error of the j^{th} parameter is $\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbf{J}^\top \mathbf{J})^{-1}]_{jj}}$. Consequently, the approximate $(1 - \alpha) \times 100\%$ confidence interval is given by $\hat{\beta}_j \pm t_{\alpha/2, n-p} \text{SE}(\hat{\beta}_j)$, valid when higher-order terms in the Taylor expansion of $f(t, \beta)$ are negligible.

1 Model 1: Exponential Sum Model

We consider the model of the form:

$$y(t) = \alpha_0 + \alpha_1 e^{-\beta_1 t} + \alpha_2 e^{-\beta_2 t} + \epsilon(t) \quad 1 \leq t \leq 60 \quad (1)$$

where α_i, β_i are parameters to be estimated. The least-squares estimators can be derived by minimizing the residual sum of squares:

$$RSS = \sum_{t=1}^{60} (y(t) - (\alpha_0 + \alpha_1 e^{-\beta_1 t} + \alpha_2 e^{-\beta_2 t}))^2 \quad (2)$$

We will use prony’s equation and osborne’s method to find the best-fitting parameters.

1.1 Osborne’s Method

Construct $G_0, G_1, G_2 \in \mathbb{R}^{n \times (n-2)}$ with $(G_0^T)_{i,i} = 1$, $(G_1^T)_{i,i+1} = 1$, $(G_2^T)_{i,i+2} = 1$.

Compute

$$U_k = G_k^T Y, \quad C_{jk} = G_j^T G_k, \quad j, k \in \{0, 1, 2\}.$$

Note that $C_{kj} = (C_{jk})^T$, ensuring symmetry between the matrices.

At each iteration:

$$\begin{aligned} G &= g_0 G_0 + g_1 G_1 + g_2 G_2, \quad A = (G^T G)^{-1}, \quad v = A G^T Y, \\ b_{jk} &= U_k^T A U_j + v^T (C_{kj} + C_{jk}) v + U_j^T A U_k, \quad B(g) = [b_{jk}]_{3 \times 3}. \end{aligned}$$

Let g' be the eigenvector of $B(g)$ corresponding to the smallest $|\lambda|$; normalize $g^{(m+1)} = g' / \|g'\|_2$, ensuring $g^{(m+1)} \cdot g^{(m)} > 0$. Stop if $\|g^{(m+1)} - g^{(m)}\| < 10^{-8}$.

Initialisation : $g^{(0)} = [5, -3, 5]^T / \|[5, -3, 5]\|_2$ was used as intial guess for the parameters.

Parameter Extraction

With $g = [g_0, g_1, g_2]^T$, solve $g_2 x^2 + g_1 x + g_0 = 0$ for roots r_1, r_2 ; then $\beta_1 = \ln |r_1|$, $\beta_2 = \ln |r_2|$.

Estimate $[\alpha_1, \alpha_2]^T = (X^T X)^{-1} X^T Y$, where

$$X = \begin{bmatrix} e^{\beta_1 t_1} & e^{\beta_2 t_1} \\ \vdots & \vdots \\ e^{\beta_1 t_n} & e^{\beta_2 t_n} \end{bmatrix}, \quad \hat{y}(t) = \alpha_0 + \alpha_1 e^{\beta_1 t} + \alpha_2 e^{\beta_2 t}.$$

1.1.1 Model parameter estimates

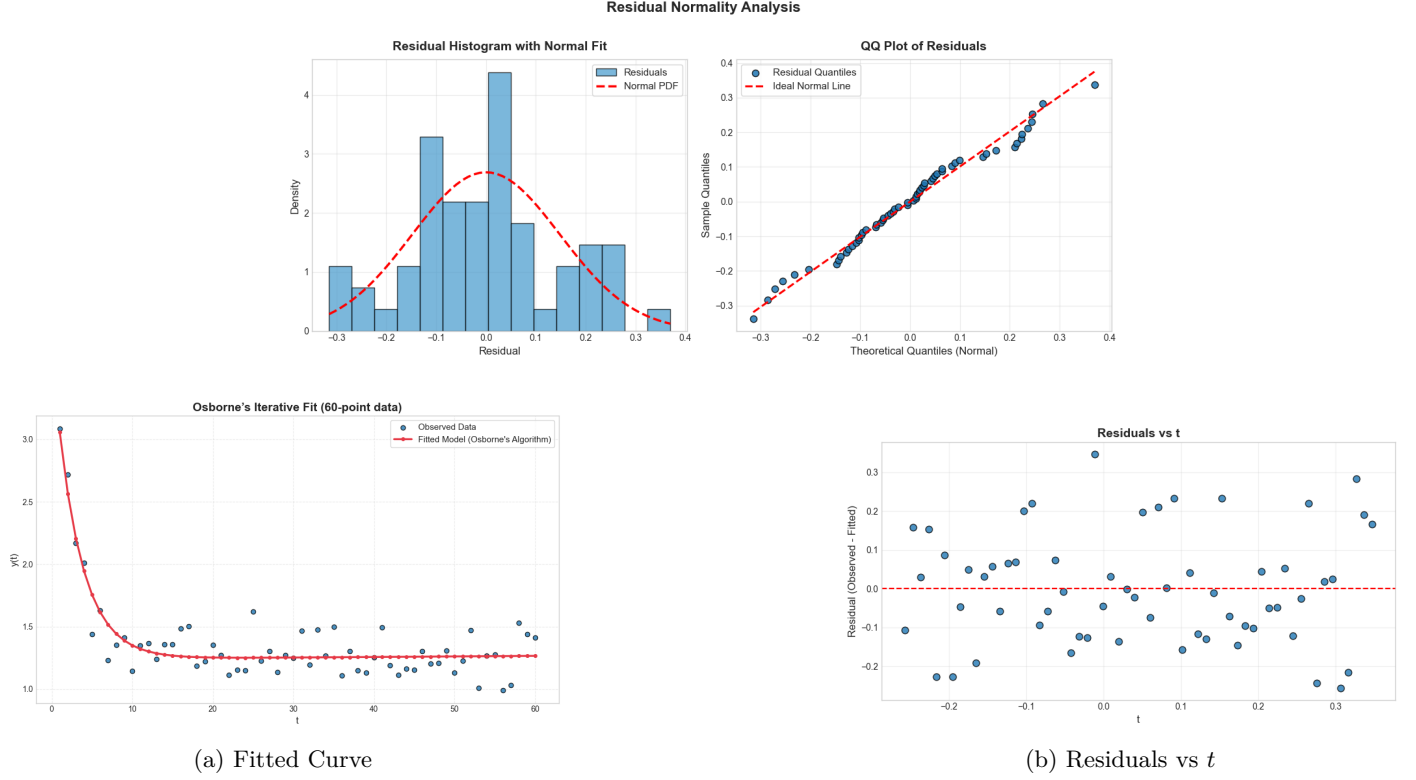
$$\begin{aligned} g &= [0.3433, -0.8133, 0.4698], \\ (e^{\beta_1}, e^{\beta_2}) &= [1.0017, 0.7294]. \end{aligned}$$

$$\alpha_0 = 1.0000, \quad \alpha_1 = 0.239046, \quad \beta_1 = 0.001739, \quad \alpha_2 = 2.491671, \quad \beta_2 = -0.315516.$$

1.1.2 Estimate of σ^2 , Plotting of Residuals and Normality check

Compute residuals $e(t) = y(t) - \hat{y}(t)$.

The unbiased estimator of σ^2 is given by $\frac{RSS}{n-m} = 0.0232$, where $m = 4$ is the number of estimated parameters and $n = 60$.



REMARKS : The residuals exhibit an approximately normal distribution. The histogram is roughly bell-shaped, with the normal curve fitting well around the center but showing slight tail deviations. The Q-Q plot further supports this, as most points align closely with the reference line, apart from minor departures at the extremes. **Overall, the residuals satisfy the normality assumption reasonably well, with only mild evidence of heavier tails.**

1.1.3 Fischer Information Matrix and 95 % Confidence Intervals

The Fisher Information Matrix (FIM) is computed as

$$\text{FIM} = \begin{bmatrix} 2880.1375 & 21716.5519 & 116.9121 & 1081.6446 \\ 21716.5519 & 212997.8906 & 103.7707 & 1661.5637 \\ 116.9121 & 103.7707 & 48.9932 & 260.8673 \\ 1081.6446 & 1661.5637 & 260.8673 & 2128.0130 \end{bmatrix}.$$

The degrees of freedom are $n - m = 56$, and the corresponding two-tailed critical value at the 95% confidence level is $t_{0.975,56} = 2.0032$.

Table 2: Parameter Estimates and 95% Confidence Intervals (Osborne's Method)

Parameter	Estimate	Std. Error	95% CI Lower	95% CI Upper
α_1	0.239046	0.060798	0.117252	0.360839
β_1	0.001739	0.006368	-0.011017	0.014496
α_2	2.491671	0.243669	2.003542	2.979799
β_2	-0.315516	0.046828	-0.409324	-0.221709

Note: The parameter α_0 was treated as a fixed constant ($\alpha_0 = 1.00$) during estimation(it was found using brute force) and thus is not included in the covariance or confidence interval computation.

2 Model 2: Rational Model

The observed data are modeled by a rational function of the form

$$y(t) = \frac{\alpha_0 + \alpha_1 t}{1 + \beta_1 t} + \epsilon(t), \quad 1 \leq t \leq 60$$

where α_0 , α_1 , and β_1 are parameters to be estimated. Here, the constant term in the denominator is fixed as $\beta_0 = 1$.

2.1 Modified Newton Gauss Method (Levenberg–Marquardt Model)

Let the residuals be defined as

$$r_i(\theta) = \frac{\alpha_0 + \alpha_1 t_i}{1 + \beta_1 t_i} - y_i, \quad \theta = [\alpha_0, \alpha_1, \beta_1]^T.$$

The objective function minimized by the Levenberg–Marquardt (LM) algorithm is

$$S(\theta) = \frac{1}{2} \sum_{i=1}^n r_i(\theta)^2 = \frac{1}{2} \|r(\theta)\|_2^2.$$

The Jacobian matrix $J = \frac{\partial r}{\partial \theta}$ has components

$$\frac{\partial r_i}{\partial \alpha_0} = \frac{1}{1 + \beta_1 t_i}, \quad \frac{\partial r_i}{\partial \alpha_1} = \frac{t_i}{1 + \beta_1 t_i}, \quad \frac{\partial r_i}{\partial \beta_1} = -\frac{t_i(\alpha_0 + \alpha_1 t_i)}{(1 + \beta_1 t_i)^2}.$$

At iteration k , the LM update is given by

$$\theta^{(k+1)} = \theta^{(k)} - (J^T J + \lambda I)^{-1} J^T r(\theta^{(k)}),$$

where $\lambda = 10^{-4}$ is the damping factor controlling the step between Gauss–Newton and gradient descent behavior. The iteration continues until $\|\theta^{(k+1)} - \theta^{(k)}\| < \varepsilon$. $\varepsilon = 10^{-8}$.

Initialization : $\theta^{(0)} = [2, -0.01, 0.5]$ is used as the initial guess for the parameters.

2.1.1 Model Parameter estimates

$$\hat{\alpha}_0 = 7.973305, \quad \hat{\alpha}_1 = 2.706315, \quad \hat{\beta}_1 = 2.328554.$$

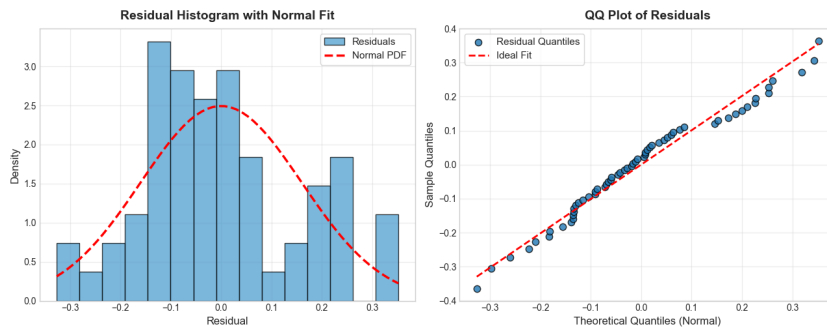
2.1.2 Estimate of σ^2 , Plotting of Residuals and Normality check

The unbiased estimate of σ^2 is computed from the residuals $r_i = r_i(\hat{\theta})$ as

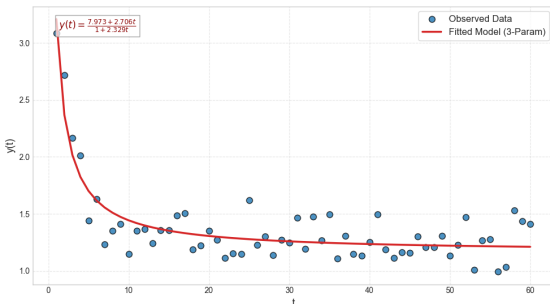
$$\sigma^2 = \frac{\text{RSS}}{n - m} = 0.0265 \quad \text{where} \quad \text{RSS} = \sum_{i=1}^n r_i^2,$$

and $m = 3$ is the number of parameters.

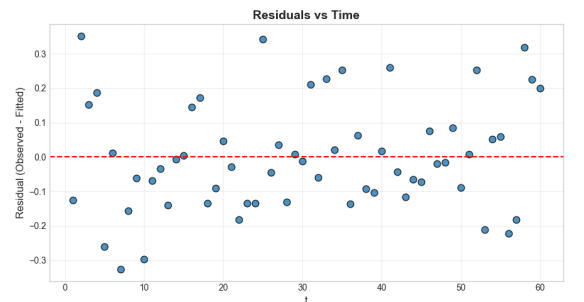
Residual Normality Analysis



Gauss-Newton Fit for Identifiable Rational Model



(a) Fitted Curve



(b) Residuals vs t

REMARKS : The residuals from the Rational Function Model show a pattern that is close to normal but with slight deviations. The histogram is moderately bell-shaped, though the fit of the normal curve suggests some skewness and mild irregularities in the tails. The Q-Q plot supports this observation—most points align well with the reference line, with small departures at both ends indicating minor non-normality. Overall, the residuals are approximately normal, and the deviations are not substantial enough to undermine the model's assumption of normality.

2.1.3 Fisher Information Matrix and 95% Confidence Intervals

$$\text{FIM} = \begin{bmatrix} 6.8254 & 25.8947 & -45.9624 \\ 25.8947 & 394.1319 & -527.0006 \\ -45.9624 & -527.0006 & 730.1159 \end{bmatrix}, \quad n - m = 57, \quad t_{0.975,57} = 2.0025.$$

Parameter	Estimate	Std. Error	95% CI (Lower)	95% CI (Upper)
$\hat{\alpha}_0$	7.973305	3.683488	0.597248	15.349362
$\hat{\alpha}_1$	2.706315	1.970409	-1.239361	6.651991
$\hat{\beta}_1$	2.328554	1.652669	-0.980858	5.637966

Table 3: Parameter estimates with standard errors and 95% confidence intervals.

3 Model 3 : Polynomial Least Squares Fit (Degree 4)

3.1 Linear Regression

We consider a fourth-degree polynomial model for the observed data (t, y_i) , where

$$y_i = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \beta_4 t^4 + \varepsilon(t), \quad t = 1, 2, \dots, n,$$

with independent and identically distributed errors. Let

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & t & t^2 & t^3 & t^4 \\ 1 & t & t^2 & t^3 & t^4 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & t & t^2 & t^3 & t^4 \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix}.$$

Then the model can be compactly expressed as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

The least squares estimator of $\boldsymbol{\beta}$ is obtained by minimizing

$$S(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}),$$

yielding

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

The fitted values and residuals are then given by

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}.$$

3.1.1 Model parameters estimates

$$\beta_0 = 2.682857, \beta_1 = -0.241045, \beta_2 = 0.013223, \beta_3 = -0.000286, \beta_4 = 0.000002.$$

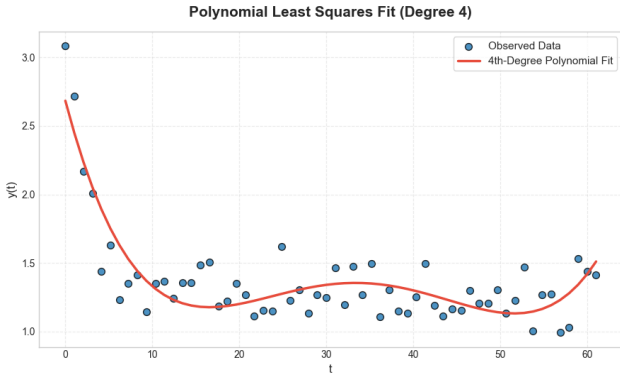
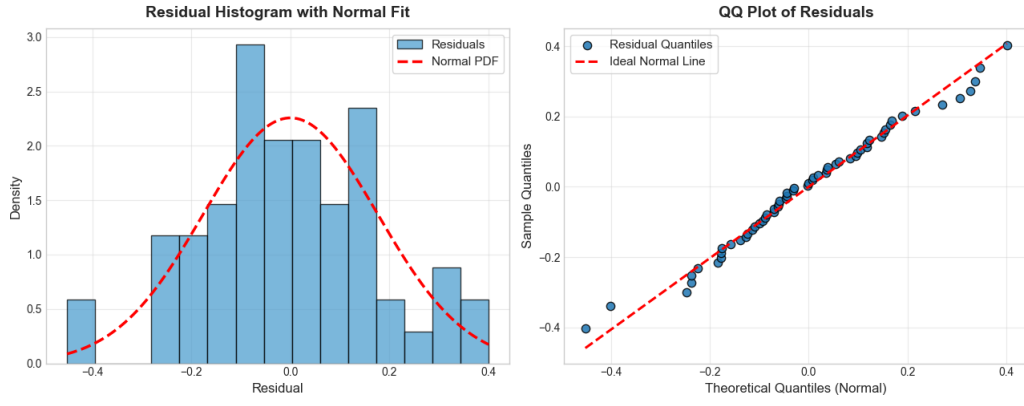
3.1.2 Estimate of σ^2 , Plotting of Residuals and Normality check

The unbiased estimate of σ^2 is computed from the residuals $\hat{\varepsilon}(i)$ as

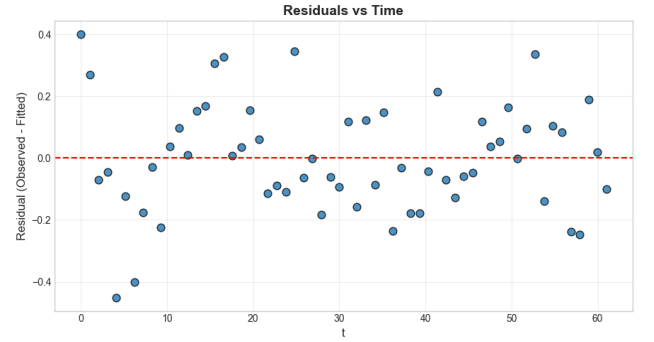
$$\sigma^2 = \frac{\text{RSS}}{n - m} = 0.034 \quad \text{where} \quad \text{RSS} = \sum_{i=1}^n \hat{\varepsilon}(i)^2,$$

and $m = 5$ is the number of parameters.

Residual Normality Analysis



(a) Fitted Polynomial Curve



(b) Residuals vs t

REMARKS : The residuals are approximately normally distributed. The histogram shows a symmetric bell shape, and the QQ plot points lie close to the reference line, indicating only minor deviations at the tails. Thus, the normality assumption holds well for the degree-4 polynomial model.

3.1.3 Fischer Information Matrix and 95% confidence intervals

$$FIM = \begin{bmatrix} 1.7900 \times 10^3 & 5.4596 \times 10^4 & 2.2391 \times 10^6 & 1.0330 \times 10^8 & 5.0832 \times 10^9 \\ 5.4596 \times 10^4 & 2.2391 \times 10^6 & 1.0330 \times 10^8 & 5.0832 \times 10^9 & 2.6055 \times 10^{11} \\ 2.2391 \times 10^6 & 1.0330 \times 10^8 & 5.0832 \times 10^9 & 2.6055 \times 10^{11} & 1.3737 \times 10^{13} \\ 1.0330 \times 10^8 & 5.0832 \times 10^9 & 2.6055 \times 10^{11} & 1.3737 \times 10^{13} & 7.3929 \times 10^{14} \\ 5.0832 \times 10^9 & 2.6055 \times 10^{11} & 1.3737 \times 10^{13} & 7.3929 \times 10^{14} & 4.0418 \times 10^{16} \end{bmatrix}$$

Degrees of freedom: $n - p = 55$

$t_{0.975,55} = 2.0040$

Coefficient	Estimate	Std. Error	95% CI (Lower)	95% CI (Upper)
β_0	2.682857	0.107284	2.467856	2.897858
β_1	-0.241045	0.024805	-0.290756	-0.191334
β_2	0.013223	0.001674	0.009868	0.016577
β_3	-0.000286	0.000041	-0.000369	-0.000203
β_4	0.000002	0.000000	0.000001	0.000003

Table 4: Parameter estimates with 95% confidence intervals.

4 Conclusion

Table 5: Model Comparison Based on Residual Analysis and Normality Tests

Model	Residual Mean	Std. Dev. ($\hat{\sigma}$)	Estimate of $\hat{\sigma}^2$	R^2	Shapiro–Wilk p	Jarque–Bera p
Exponential	-6.8×10^{-6}	0.148	0.0232	0.836	0.204	0.392
Rational	9.3×10^{-15}	0.160	0.0265	0.797	0.152	0.401
Polynomial (deg 4)	-4.3×10^{-10}	0.177	0.0335	0.753	0.720	0.976

Model Comparison and Conclusion: All three models—exponential, rational, and polynomial—were fitted to the given dataset, and their performance was evaluated using the residual variance ($\hat{\sigma}^2$), coefficient of determination (R^2), and residual normality tests (Shapiro–Wilk and Jarque–Bera). The exponential model achieved the lowest residual variance ($\hat{\sigma}^2 = 0.0232$) and the highest coefficient of determination ($R^2 = 0.836$), indicating the best overall goodness of fit. Although the polynomial model exhibited the highest normality p -values, its residual variance ($\hat{\sigma}^2 = 0.0335$) and R^2 value (0.753) were inferior, suggesting that it overfits local fluctuations without improving overall predictive accuracy. The rational model showed intermediate performance ($\hat{\sigma}^2 = 0.0265$, $R^2 = 0.797$) but slightly weaker normality.

Hence, considering both the model fit and residual behavior, the *exponential model* is identified as the most suitable representation of the dataset.

5 References

Stat 5102 Notes: Fisher Information and Confidence Intervals Using Maximum Likelihood by Charles J. Geyer, *Gauss-Newton / Levenberg-Marquardt Optimization* by Ethan Eade, *Wikipedia : Fisher information, Shapiro–Wilk test, Jarque–Bera test*